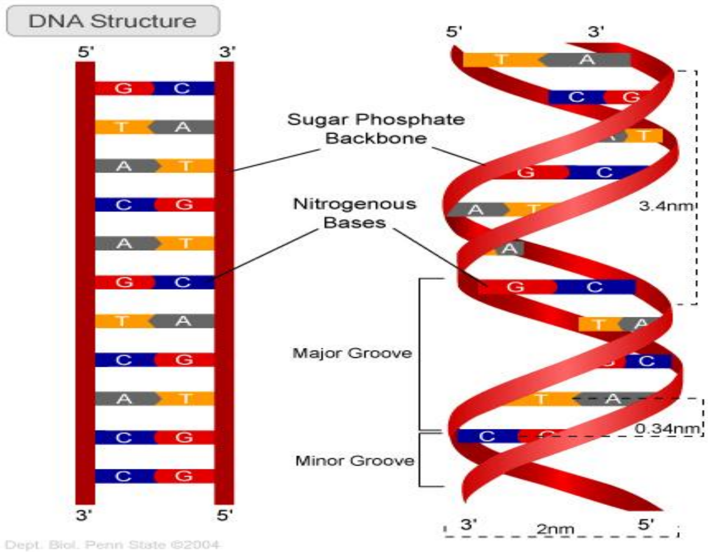
L no 4 Molecular basis of Inheritance

1. Draw a neat well labeled diagram of structure of DNA(3M)



Answer in detail.

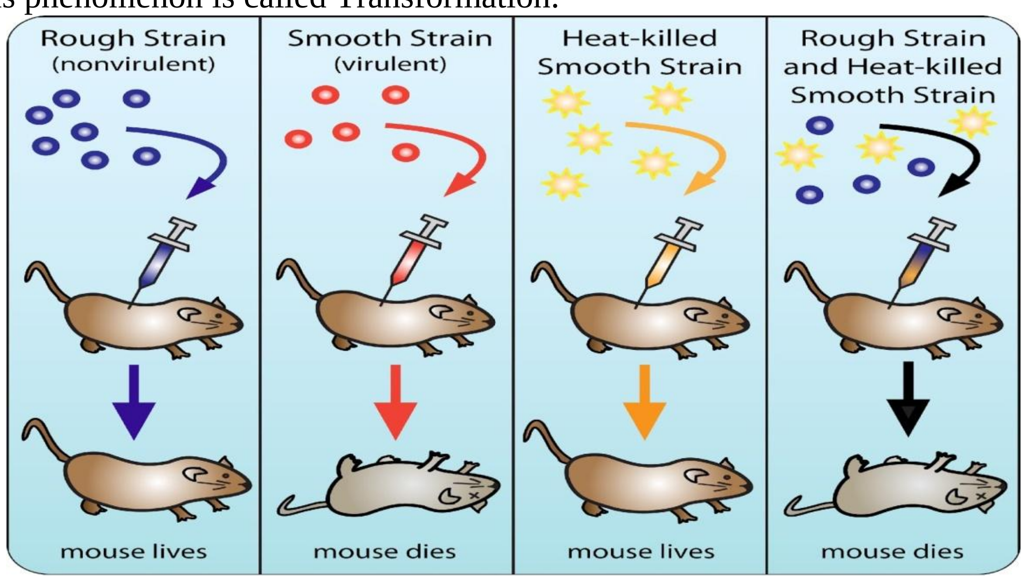
Q. Explain Griffith's experiment how he proved DNA as genetic material. (3M)

Ans:*1.Diplococcus pneumonia* bacteria include two strains, a ***virulent*** **S** strain with a ***S***mooth glycoprotein coat that kills mice (left), and a ***non-virulent*** **R** Rough strain that does not cause disease Heating destroys the virulence of **S** strain

2. In the critical experiment, **Frederick Griffith** (**1928**) mixed *heat-killed***S** with *live* **R** and injected the combination into mice: the mouse died. The dead mouse's tissues were found to contain *live* bacteria with *smooth coats* like **S**. These bacteria were subsequently able to kill other mice, and continued to do so after several generations in culture.

3. Griffith concluded that *something* in the heat-killed **S** bacteria '***transformed'*** the *hereditary* properties of the **R** bacteria.

This phenomenon is called Transformation.



Q. Describe Harshey& Chase experiment to prove that DNA is genetic material.(2M)

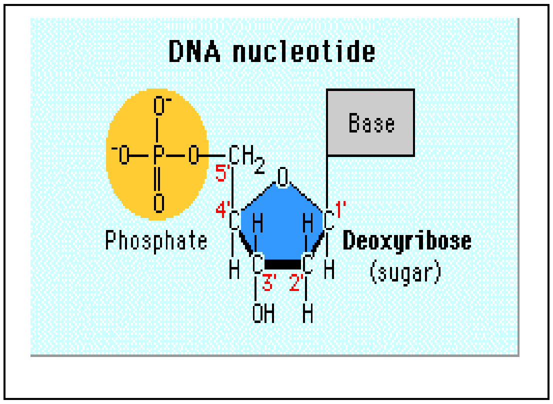
Ans: 1. Harshey & Chase worked to discover whether the genetic material of infecting virus to bacteria was DNA/ protein

2. They used radioactive phosphorus P^{32} in a medium for some viruses and radioactive sulphur $S^{35}$for some others.
3. DNA contains phosphorus and not sulphur while proteins contain sulphur and not phosphorus. So some viruses were with radioactive DNA while some with radioactive protein.
4. Both were allowed to infect *E-coli* bacteria. As the infection proceeded the viral coat were removed with the help of process called centrifuge.
5. Bacteria which were infected by viruses with radioactive DNA were radioactive while the viruses which were infected with radioactive protein were not radioactive

This indicates that DNA enters the bacterial cells and not the proteins**Therefore DNA is a genetic material.**

Q. What will be length of eukaryotic DNA segment having 10 pairs of nucleotides?(1M)
Ans: The length of eukaryotic DNA segment having 10 pairs of nucleotides will be 3.4^0 or 3.4nm

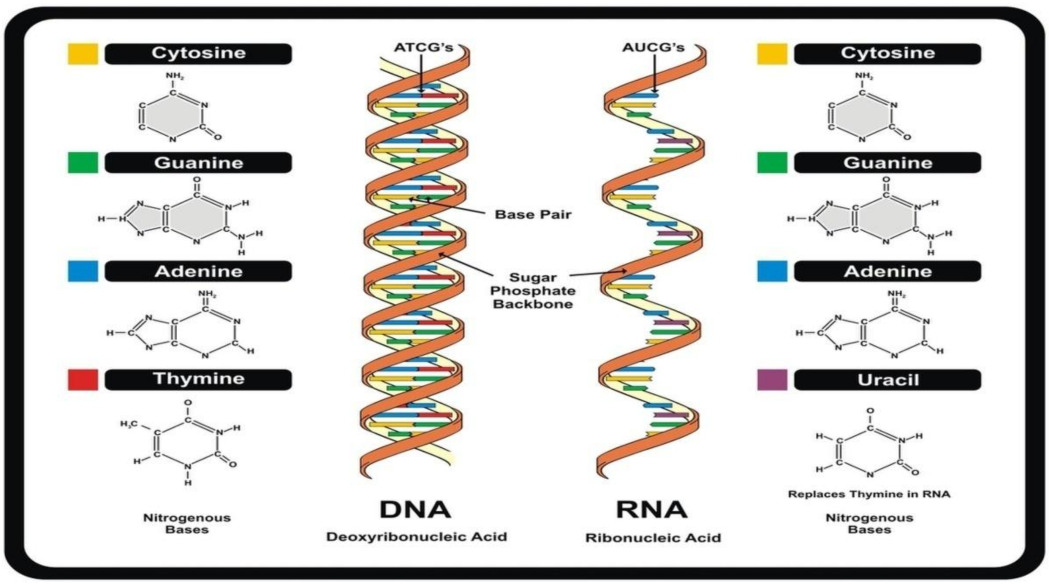
Q.Describe chemical components of DNA.(3M)(*all components necessary for full marks***)**
Ans: Nucleotides are the structural units of nucleic acid. Each nucleotide has 3 components.
Sugar: It is a pentose sugar called deoxyribose or ribose. It is 5 carbon compounds & has pentagonal ring structure.
Phosphate Group: It is derived from phosphoric acid. H_3PO_4.
Nitrogen Bases: 1) There are 4 bases in DNA viz. Adenine (A), Guanine (G), cytosine(C), Thymine (T).
2) A& G are double ring purines while G & C are single ring pyrimidines.
3) The number of purines is always equal to pyrimidines.
A $=$ T andG $\equiv$ C
4) Hence ratio of Purine: pyrimidine is always constant.Hence
Nucleotide= nitrogen base + Pentose sugar+ phosphate group
Nucleoside=nitrogen base + Pentose sugar
Therefore**, Nucleotide=Nucleoside +Phosphate**



Q. Distinguish between DNA and RNA

Table 11.1: Differences between DNA and RNA

DNA	RNA
1. DNA is found in the chromosomes of the nucleus and is mainly concentrated in the nucleus only.	1. RNA is mainly concentrated in the cytoplasm, although it also occurs in the nucleolus and, to some extent, in the nucleoplasm and is associated with the chromosome.
2. DNA is a double-stranded structure. The two strands are coiled spirally in opposite directions.	2 RNA molecules are single-stranded. The strand in some tRNA is the duplex form and is connected by hydrogen bond.
3. The sugar molecule in DNA is deoxyribose.	3. The sugar molecule in RNA is ribose.
4. The four nitrogenous bases found in DNA are—Adenine, Guanine, Cytosine and Thymine.	4. The four nitrogenous bases found in RNA are—Adenine, Guanine, Cytosine and, in RNA, Thymine is replaced by uracil.
5. The purine and pyrimidines of DNA are found in equal proportions.	5. It is not necessary.
6. DNA is the hereditary material and the information for different cell activities and life processes are encoded in DNA molecule.	6. Except some viruses, RNA participates in protein synthesis.
7. The base composition in DNA is A/T = G/C = 1.	7. Not so in RNA.



Q. Explain structure of DNA(7M) (*3M diagram, 4M any 4 subpoints with explanation* **)**

Ans: In 1953 Watson & crick propsed DNA structure based on X-ray crystallogrophy.

I) DNA Molecule as double helix:

1.DNA molecule consists of 2 long strands coiled around a common, imaginary \, central axis to form a double helix.

2.It looks like twisted ladder and shows major & minor grooves.

II) Structure of each strand

1.Each strand of DNA consist of number of nucleotides

2. Each nucleotide is made up of deoxyribose sugar, phosphate group and a nitrogen base

III) Complementery base Pairing

1. Adenine always pair with Thymine with double hydrogen bond while Guanine always pair with Cytosine with triple hydrogen bond.

A=T G ≡ C.

Hence the strands are also called as complementary strand.

IV) Purine Pyrimidine Ratio

1. Due to complementary base pairing total number of purine bases is always equal to total number of pyrimidine bases (1:1 ratio)
2. This is called Chargaff's rule
3. It may be represented as follows:-

   A+G=T+C OR $\frac{A+G}{T+C} = 1$

V) Polarity of strands

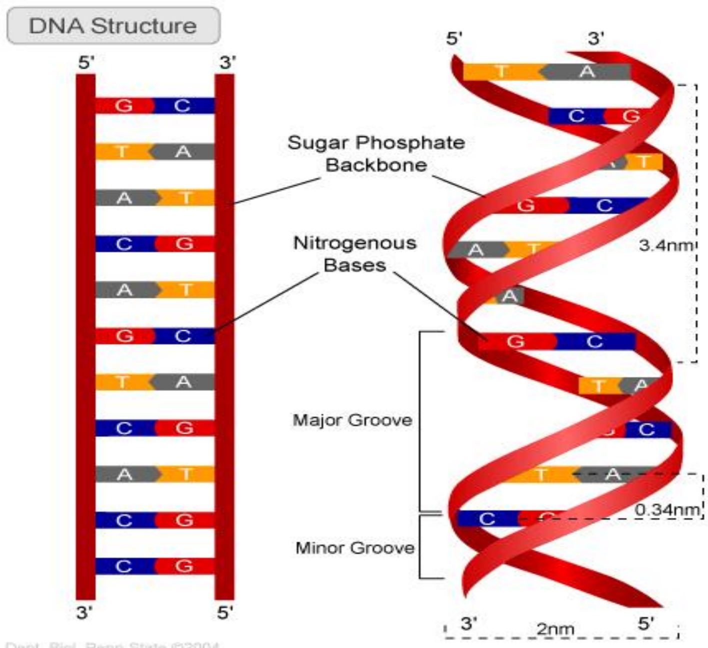
1. 1 end of strand is 5' and the other end is 3'
2. At 5' end free phosphate group while at 3' end is OH group
3. Both the strands run antiparallel to each other

VI) Major & Minor Grooves

1. The strand of DNA undergo right handed coiling around imaginary axis.
2. This coiling results in formation of major and minor groves

VII) Dimensions

1. Diameter of DNA=2nm
2. Distance between 2 major and minor grooves is 3.4nm or $34A^0$
3. There are 10 base pairs in 1 complete spiral.



Q. Explain replication of eukaryotic DNA

Ans: <u>DNA Replication</u>

The process in which a DNA molecule produces an exact copy or replica of itself is known as replication of DNA.

<u>Mechanism of DNA Replication</u>

a) Activation of nucleotides

i) Nucleotides are found in nucleoplasm.

ii) Activation occurs in presence of the enzyme phosphorylase and the process is called activation of nucleotides.

iii) (dAMP+ATP) $\longrightarrow$ (dATP+AMP)

b) Origin / Initiation points

i) Replication of DNA begins at certain specific sites situated on the molecule.

ii) Such sites are called as origin points or initiation points.

c) Unwinding of DNA strands

i) After the activity of endonuclease enzyme Y shaped structure called replication fork appears on DNA strand.

ii) Separated strands are prevented from coiling by SSBP i.e. (Single strand DNA Binding protein)

d) ***Synthesis of new strand***

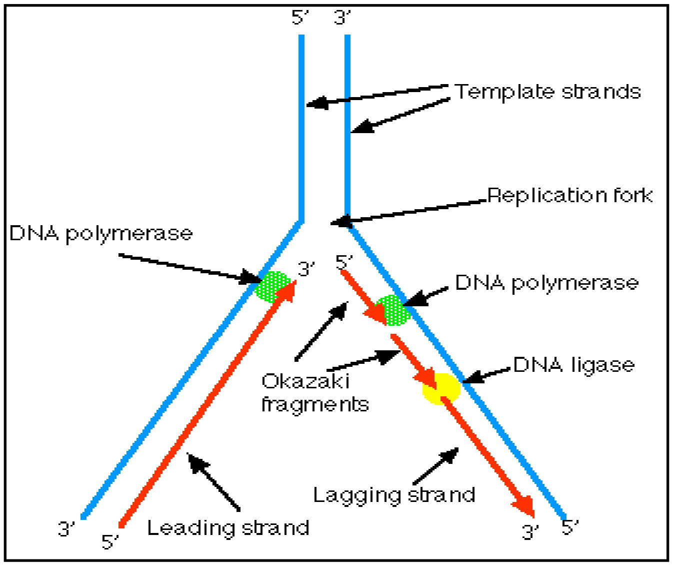
i) for synthesis of complementary new strand, each separated strand acts as template/ mould with the help of RNA primer

ii) RNA primer attach itself to 3' end of template strand and attracts the new nucleotides from nucleoplasm: appropriate nucleotides are attached by H-band to their respective complementary bases on the strand. Synthesis of new strand occurs in presence of enzyme DNA polymerase.

e) ***Leading and Lagging strand***

i) The strand which opens from 3'-----5' is continuous called as leading starndit is constructed at the faster speed.

ii) While other strand 5'-----3' is discontinuous called as lagging strand



1. **Describe characteristics of genetic code.(any4 subpoints with explanation)(2marks)**

Ans: 1) Commaless: Genetic code is triplet and commaless, i.e. on the mRNA strand the triplet codons are arranged one after the other without any gap or space
2) Non-ambigous: Each codon will specify a particular aminio acid**.**
3) Degenerate: the genetic code is degenerate as 61 codons are available 20 amino acids. 2 or more codons can specify the same amino acid. Hence it is said to be degenerate. E.g. GGG, GGA, GGC, and GGU code for glycine.
4)Polarity: It can be read only in 5' ----3' direction of mRNA. The genetic code has start and stop signals. The AUG acts as start or initiation codon, whiler UAG, UAA and UGA serve as stop codons or termination codons.
5) Universal: The genetic code is universal i.e. it is similar in all the organisms, from simple bacteria to complex organism. (except mitochondria/ yeast & mycoplasma)

Q. What is RNA? Describe various types of RNA. (7M)
(def:1M, each type with explanation and function :3M, all3 diagrams:3M) (Note t-RNA any 1 diagram required)
AnsDefinition:***RNA is single stranded polynucleotide molecule.***
There are 2 types of RNA. Viz.
Genetic RNA and non- genetic RNA
RNA acts as genetic material in few viruses
Types of non-genetic RNA

I) m-RNA / Messenger RNA

1. It is always simple &straight without any fold.

2. About 3-5% of total cellular RNA is of this type
3. It has molecular weight 5, 00,000 Daltons.
4. **Function:** It carries message for protein synthesis from DNA to ribosome.

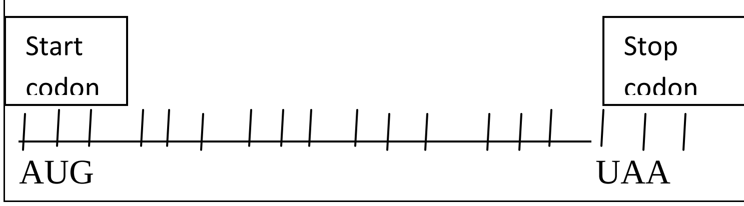
II) <u>r-RNA/ Ribosomal RNA</u>

1. It is a single stranded structure which is variously folded upon itself.
2. In the folded regions it may show pairing between complementary bases
3. About 80% of total cellular RNA is ribosomal RNA.
4. Molecular weight ranges between 40,000-100,000 Daltons.
5. **Function:** It provides proper binding site for m-RNA
   It orients the m-RNA molecule in such a way that all the codons are properly read.

III) t-RNA/ transfer RNA

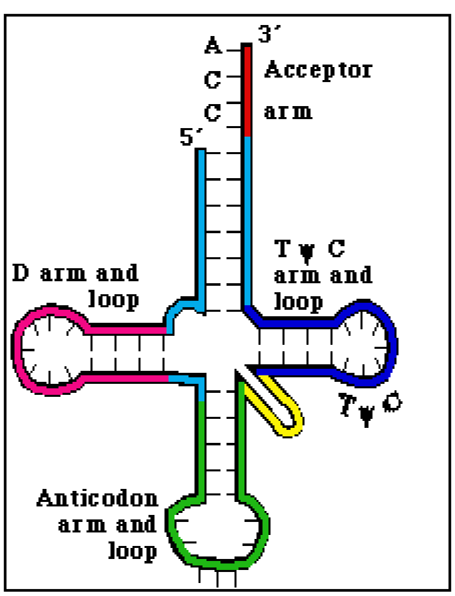
1. It is also single stranded.
2. About 10-20% of total cellular RNA is of this type.
3. Molecular weight ranges between 23,000-30,000 Daltons.
4. It has 2 structures viz. Hair pin model and clover leaf model.
5. **Functions:** It carries specific type of amino acid at CCA end to the ribosome during protein synthesis.
   It places the required amino acid properly in the sequence and translates the coded message of mRNA in terms of amino acid.

mRNA



rRNA

t-RNA



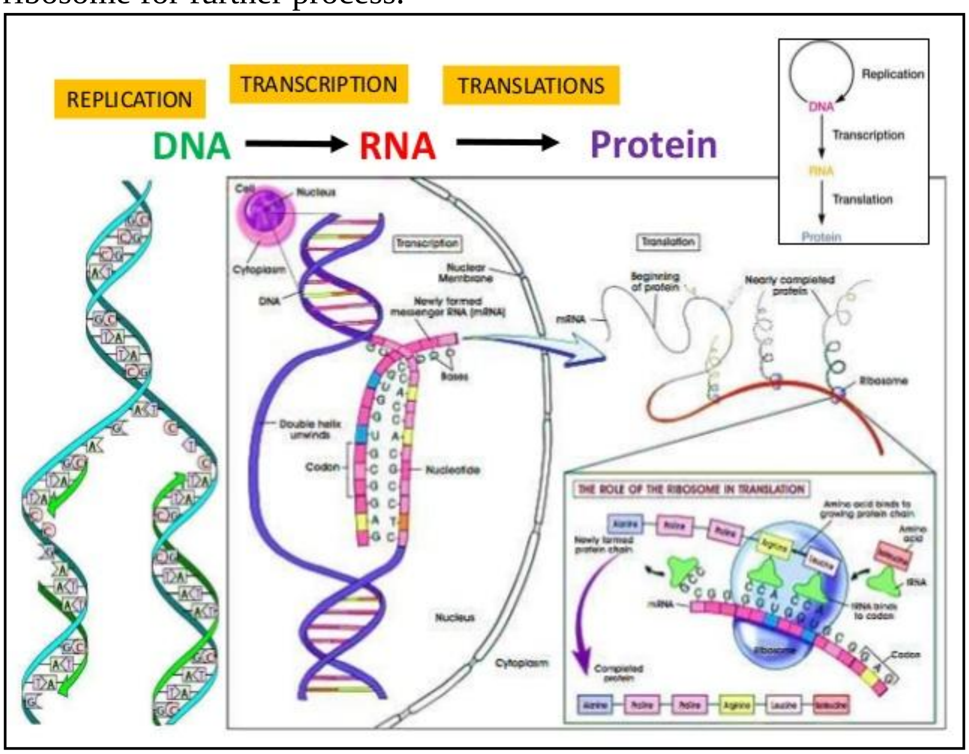
2. Explain transcription process in detail. (7M)
(Central Dogma:1M, Def:1M, Explanation:2M, Diagram:3M)

Ans: <u>Central Dogma of Protein Synthesis</u>

DNA $\xrightarrow{\text{Transcription}}$ **mRNA** $\xrightarrow{\text{Translation}}$ **Protein**

<u>Transcription</u>

1. Transcribe means copy in writing.
2. **Def:**The process of formation of mRNA on the DNA strand is called transcription.
3. In this process, genetic information from DNA is copied into RNA. It is the first in protein synthesis which takes place in the nucleus.
4. It takes place in presence of enzyme RNA polymerase. This enzyme also catalyzes the polymerization in only 1 direction. i.e. 5’ $\rightarrow$ 3’.
5. The transcription occurs in definite part of DNA.
6. The 2 strand of DNA separate by uncoiling & temporary breakage of hydrogen bond occurs.
7. One of these separated strand is selected as active template or antisense strand for synthesis of mRNA. Template strand orient in 3’ $\rightarrow$ 5’ direction or polarity.
8. The other strand which has polarity (5’ $\rightarrow$ 3’) & the sequence same as RNA, this strand is not involved in RNA synthesis and is called coding or sense strand or mRNA like strand.
9. The exposed nitrogen bases of template strand attract the complementary ribonucleotides having AUGC from the surrounding.
10. The nucleotides link with each other & form mRNA.
11. The formation of mRNA always start from initiation codon AUG at 5’ end & ends in any 1 of the stop codon UAA, UAG, UGA at 3’ end.
12. The newly formed mRNA separate from DNA strand. After separation of mRNA the 2 strand of DNA rewind and form double helix.
13. After formation, the mRNA enters the cytoplasm through nuclear pore & attaches to ribosome for further process.



6. **Explain translation in detail.(7M)**
   (Def:1M, Explanation :3m, Diagram:3M)

Ans: **Translation** *It is the process in which the sequence of codons on the mRNA strand is used and accordingly the amino acids are joined to each other to form a polypeptide chain that makes protein.*

This process involves following steps;

A) **Activation of amino acids & formation of AA-tRNA complex**.
In presence of an enzyme amino acyl tRNAsynthetase the amino acid molecule is activated & then each amino acid is attached to specific tRNA molecule at 3’ CCA end to form amino acyl tRNA complex. *The reaction needs ATP. This process is called charging of tRNA or aminoacylation of tRNA*

B) Formation of polypeptide chain
It is actual translation which involves following steps:

i) Initiation

1. It begins with formation of initiation complex which require the mRNA having codons for a polypeptide, the smaller 30S and larger 50S subunits of ribosome, the initial AA-tRNA complex and ATP & GTP act as source of energy.
2. The process starts with binding of mRNA on smaller 30 S subunit of ribosome.
3. The start codon AUG is positioned properly.
4. The AA-tRNA complex now gets attached to start codon AUG.
5. Small & large unit of ribosome join to form 70S ribosome.
6. The ribosome has 3 sites: a) amino acyl site (A)
   b) peptidyl site (P)
   c) Exit site (E)
7. Only the AA-tRNA complex binds at P site directly while all the other incoming tRNA complexes get attached first at A site & then are shifted to P site.

ii) **Elongation**

1. After initiation the process of elongation starts.
2. The elongation activity is catalyzed by enzyme peptidyltransferase.
3. Each tRNA complex brings a specific amino acid.
4. During elongation, the ribosomes moves along the mRNA in stepwise manner from staert to stop codon ahead each time. This movement is called translocation.
5. In every step of translocation 1 amino acid is added in polypeptide chain casing elongation.

iii)**Termination:**

1. Elongation continues until the ribosomes adds the last amino acid coded by the mRNA.
2. When the mRNA reaches to the last termination codon i.e. either UAA, UAG or UGA termination occurs,
3. After termination the smaller 30S and larger 50S ribosome get separated from each other.
4. The energy required for protein synthesis activity is fulfilled by ATP & GTP.

C) Polysomes/ Polyribosomes
In order to increase cellular efficiency of protein synthesis many ribosomes may bind to the mRNA strand and form the polypeptide chains for synthesis of protein

molecule. Such a structure with many ribosomes is called polysomes/ polyribosomes.

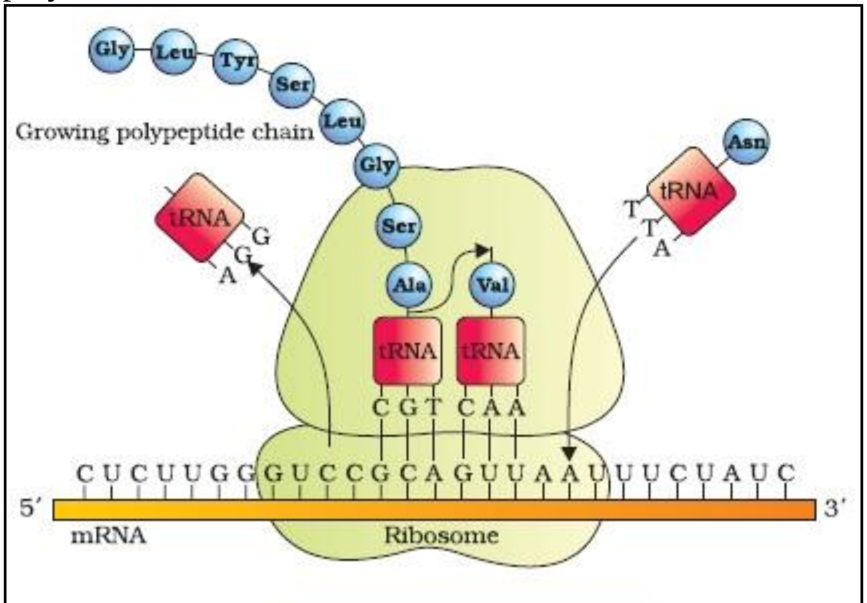


Figure 6.13 Translation

7. Describe structure of operon in detail. (3M)

Ans:1. The cluster of genes with related functions is known as **operon.**

2. Each operon is a unit of gene expression & regulation which includes the **structural genes**& their control elements (**promoter & operator)**

3. The **structural genes** code for proteins, rRNA and tRNa required by the cell.

4. **Promoters** are the signal sequences of the DNA that start RNA synthesis. These are the sites where the RNA polymerases are bound during transcriprtion.

5.The**operators** are present between the promoters & structural genes. The repressor protein binds to the operator region of the operon.

7. Explain Lac operon in detail.(7M)***(Diagram:3M, Explanation:4M)***

Ans: The metabolism of lactose in cell requires 3 enzymes: a) permease
b) β galactosidase
c) Transacetylase

Functions of these enzymes

Permease: It is needed for entry of lactose in the cell.

β galactosidase: It brings about hydrolysis of lactose into glucose &galactose

Transacetylase: It transfers acetyl group from Acetyl coA to β galactosidase

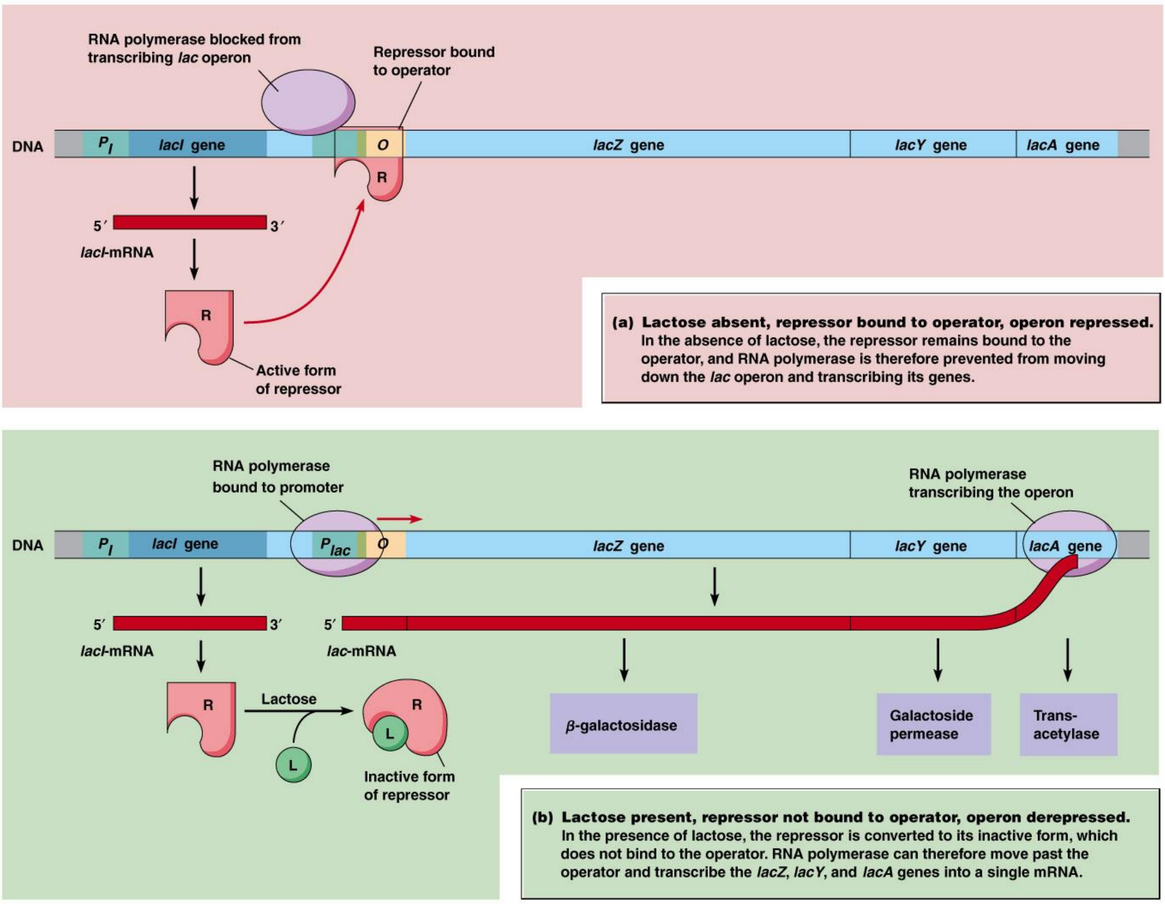
LAC OPERON

1. The Lac operon has 3 sites: a) promoter site (P)
b) Operator site (O)
c) Regulatory site (R)

Besides it has 3 structural genes viz. z, y, a. They code for following enzymes.
z= β galactosidase
y= permease a= transacetylase

2. When the cell is using its normal energy source glucose, the ‘i’ gene transcribes a repressor mRNA after translation of which a repressor protein is produced.
3. It binds the operator region of the operon and prevents the RNA polymerase from transcribing the operon.

4. In absence of glucose if lactose is available as energy source for the bacteria then following events occur in the cell.
5. The lactose enters the cell as a resultof the activity of permease enzyme.
6. It acts as inducer and interacts with the repressor to inactivate it.
7. As the repressor is inactivated the rNA polymerase bind itself to operator site and transcribe the operon to produce lac mRNA which enables formation of all the required 3 enzymes needed for lactose metabolism.
8. This regulation of lac operon by the repressor is an example of negative control of transcription initiation.



(a) Lactose absent, repressor bound to operator, operon repressed. In the absence of lactose, the repressor remains bound to the operator, and RNA polymerase is therefore prevented from moving down the *lac* operon and transcribing its genes.

(b) Lactose present, repressor not bound to operator, operon derepressed. In the presence of lactose, the repressor is converted to its inactive form, which does not bind to the operator. RNA polymerase can therefore move past the operator and transcribe the *lacZ*, *lacY*, and *lacA* genes into a single mRNA.

Q What is anticodon.(1M)

Ans. Anticodon is a triplet of nucleotides present on the anticodon loop of t-RNA which is complementary to the codon of m-RNA.

Q Give the names and functions of enzymes involved in lactose metabolism in E-coli.(3M)

Ans;The metabolism of lactose in cell requires 3 enzymes: a) permease
b) βgalactosidase
c) Transacetylase

Functions of these enzymes

Permease: It is needed for entry of lactose in the cell.
βgalactosidase: It brings about hydrolysis of lactose into glucose &galactose
Transacetylase: It transfers acetyl group from Acetyl coA to β galactosidase

Q. Explain Chargaff's rule.(1M)

1. Due to complementary base pairing total number of purine bases is always equal to total number of pyrimidine bases (1:1 ratio)
2. This is called Chargaff's rule
3. It may be represented as follows:-

$A+G=T+C$ OR $\frac{A+G}{T+C} = 1$